

The Arcade Blogger

Classic

As Atari's best-selling arcade game of all time, *Asteroids* was literally a game changer. Released in December 1979, it was responsible for catapulting Atari into mainstream public consciousness. This was the game that single-handedly broke the stranglehold that *Space Invaders* had on the video game world.

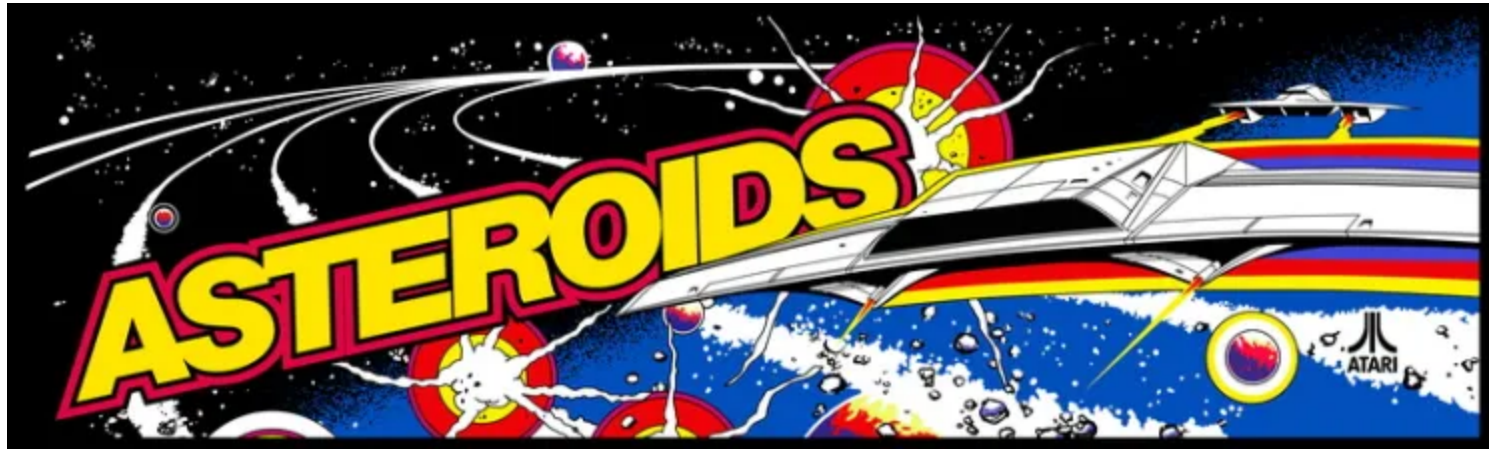
Although created by Atari developer Ed Logg (whose credits include *Super Breakout*, *Video Pinball*, *Asteroids*, *Centipede*, *Millipede*, *Xybots*, *Gauntlet*, *Space Lords* and *Steel Talons*) its lineage is rather interesting.

I guess the way I describe it is that I'm the father of Asteroids. Ed Logg is the mother of Asteroids, because he had to live with it for nine months and deliver finished product. All I had to do was to provide a seed.

Those are the words of Lyle Rains, Ed's boss at the time. Rains had been thinking about an older game developed in-house at Atari that involved two players trying to shoot each other (*Computer Space* style) with a big rock between them. The game simply wasn't fun, and all players wanted to do was shoot the rock, but of course they couldn't. The game was eventually shelved.

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Rains and Logg recall that the brief was very simple. There was no great detail at that stage, just the start of a broader idea. They did however quickly agree on the name – *Asteroids*.



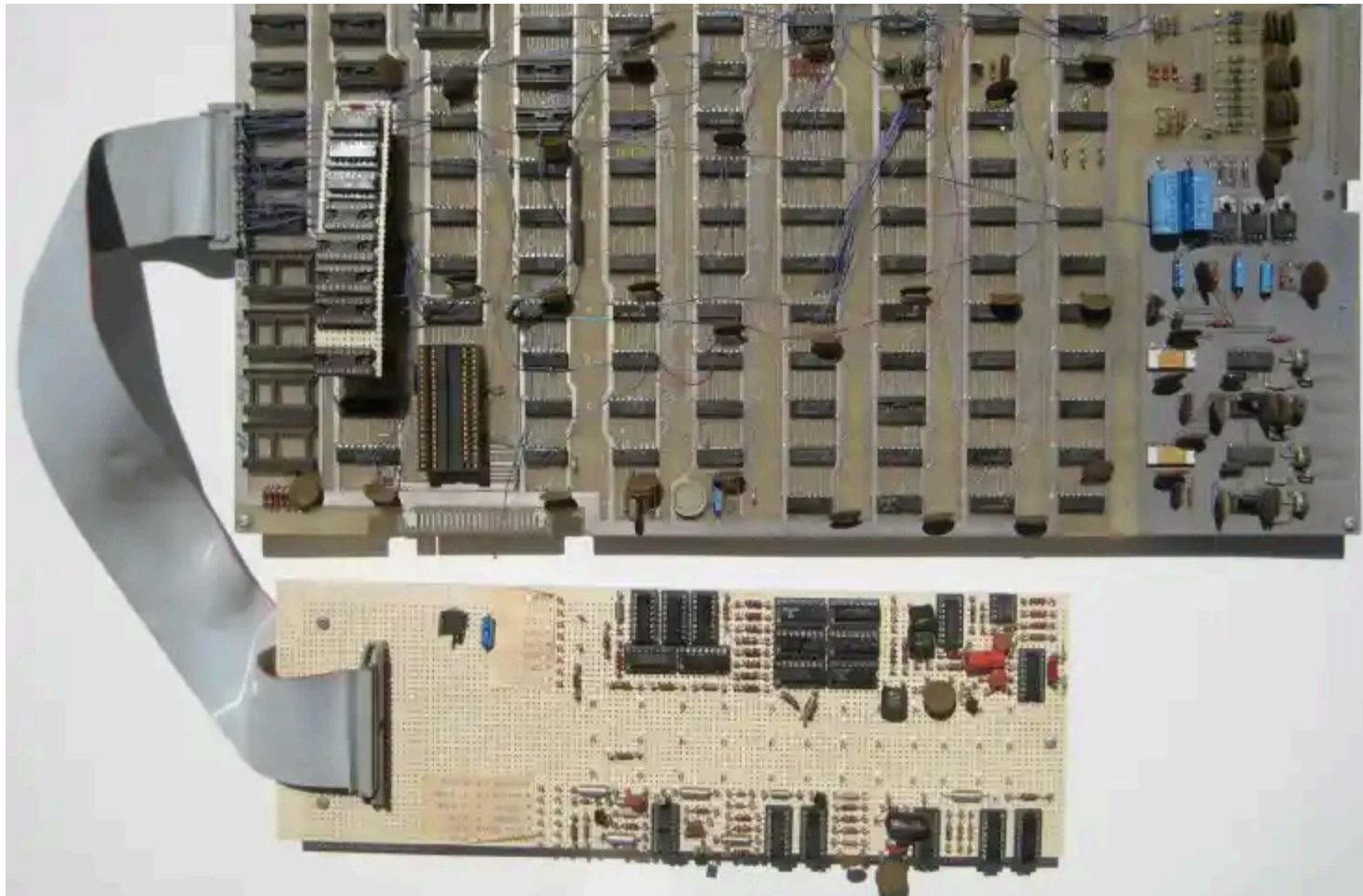
The Asteroids marquee

Inspired, Ed got to work. He pushed for the game to be developed on Vector hardware. Going against Rains' wishes for a raster title, Ed envisioned more accuracy and definition from the same (then new and innovative) hardware used to develop *Lunar Lander*. He quickly tracked down Atari's hardware engineer Howard Delman, who listened intently to Ed's idea. Recalling another game that went nowhere, he pulled out a vector board and plugged it in. The game was called *Cosmos* and later, *Planet Grab*, and involved two players flying around trying to claim and steal planets from each other.

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Within a couple of days, Delman had developed a basic hardware development kit for Logg to work on. In fact it was a modified *Lunar Lander* board:

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The original Lunar Lander vector PCB board used to design and create Atari's Asteroids. Note the small daughter board on the left containing the additional memory required to run the new Asteroids code. The blue jumper wires were added to modify the hardware, and the satellite board connected below was hand-built to produce the sounds for the game. Once complete, a single board incorporating everything was developed. Photo:

Howard Delman

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Amazingly, the finished game is almost exactly as he describes it here:

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ENGINEERING TEAM: Ed Logg / Howie Delman / Paul Mancuso

BRIEF DESCRIPTION OF GAME PLAY: The object of the game is to destroy the asteroids and saucers. Shooting a large asteroid breaks that asteroid into 2 medium sized asteroids. Shooting one of those pieces breaks it into 2 small asteroids. Shooting (over)

PLAY FEATURES TO EMPHASIZE: Scoring: Small rock = 100, medium rock = 50, large = 20 large saucer = 200, small saucer = 1000. Every 10,000 points you will get an extra ship. Using hyperspace is risky because there is a probability of destruction upon reentry.

PLAYER OPTIONS (if any): Controls: 2 (left + right) rotate buttons, hyperspace buttons, thrust button and fire button

OPERATOR OPTIONS:

Game Times: None

Extended Play (describe score levels, amount of time, etc.):

None

Coinage: Free play, 1 coin 2 plays, 1 coin 1 play, 2 coins 1 play

Middle coin mech = 1 or 2 mins; right coin mech = 1, 4, 5 or 6 coins

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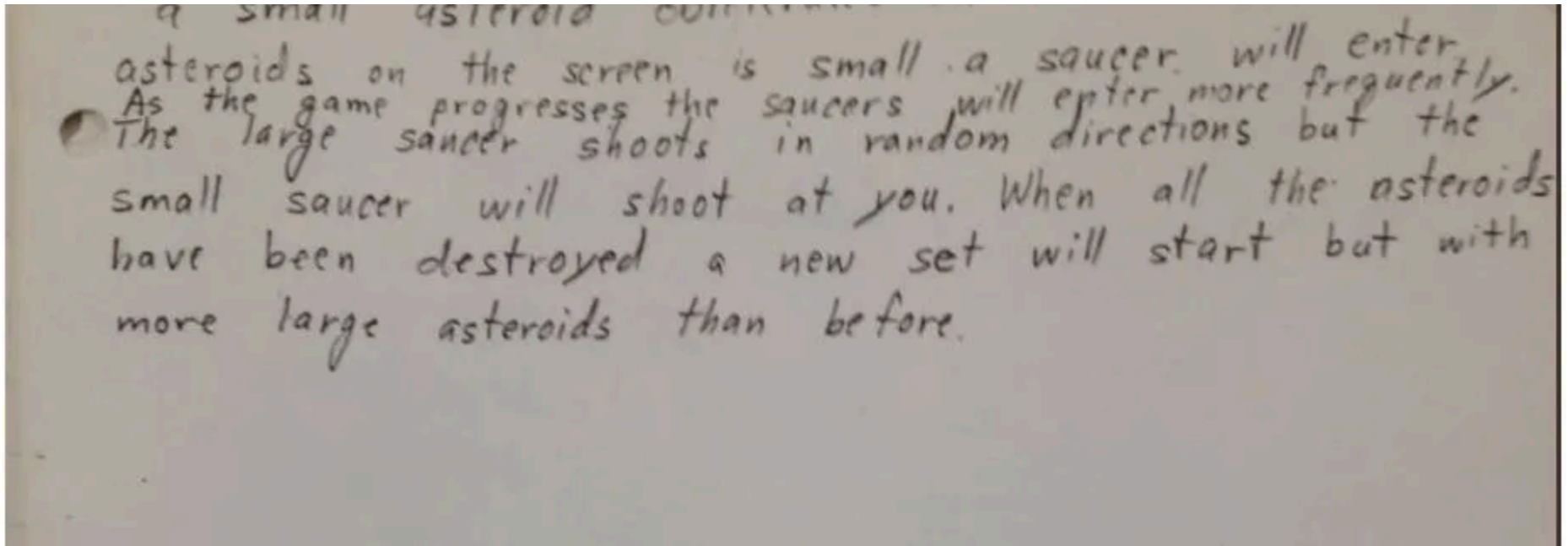
OVERALL DIMENSIONS: Height _____ Width _____ Depth _____

MONITOR SIZE: X-Y monitor (standard size)

OTHER FACTORS: If the players score is one of the ten best, he/she
and the players initials,
to enter his/her initials. The ten best scores are displayed
in attract mode.

Credit: Museum of Play, Rochester NY

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Credit: Museum of Play, Rochester NY

The object of the game is to destroy the asteroids and saucers. Shooting a large asteroid breaks that asteroid into two medium-sized asteroids. Shooting one of those pieces breaks it into two small asteroids.

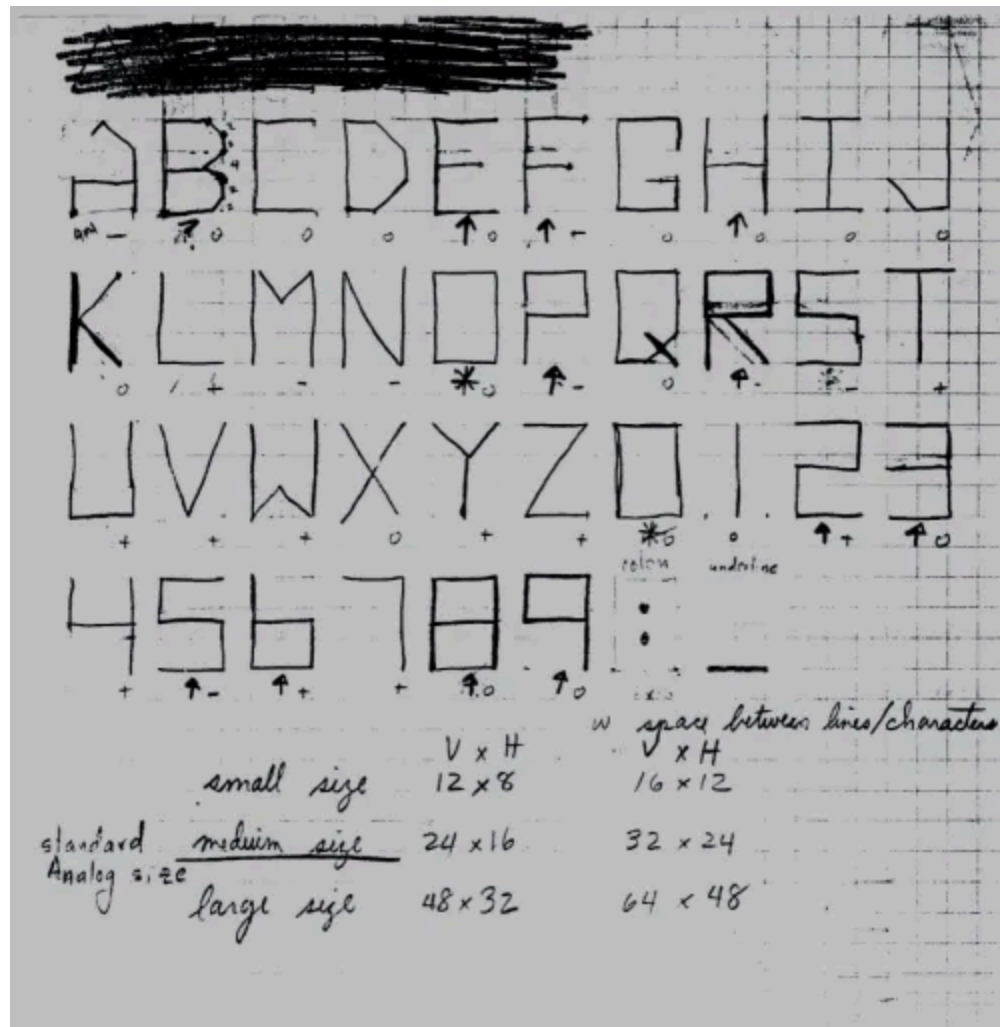
Another interesting development document can be viewed [here](#).

Logg recalls the intensity of the development process:

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going to look like before I even get there.



The Asteroids font. Designed and hand-drawn by Ed Logg here on this document from 1979. (Courtesy of Ed Logg)

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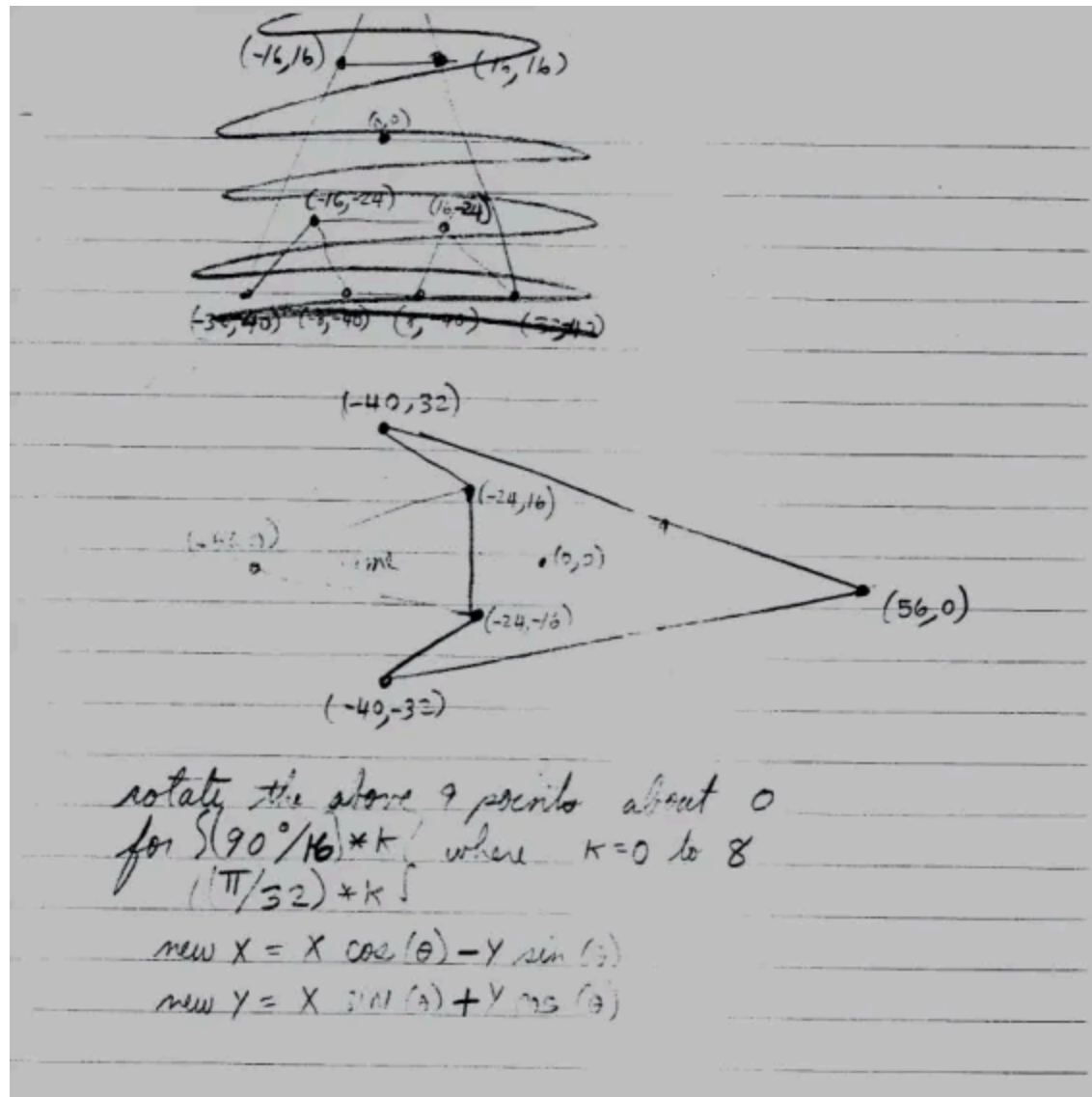
this resulted in too many accidental deaths and a feeling that the player wasn't fully in control of the craft. Settling on the gentle mid-point glide effect of the ship gave players just the right balance between control and danger.



Edd Logg pictured circa 1982

A side-effect of the vector hardware used, also delivered some noticeable features in *Asteroids*. As well as the simple bleakness of the white on black vectors, Logg makes reference to the glow that follows the player's ship around the

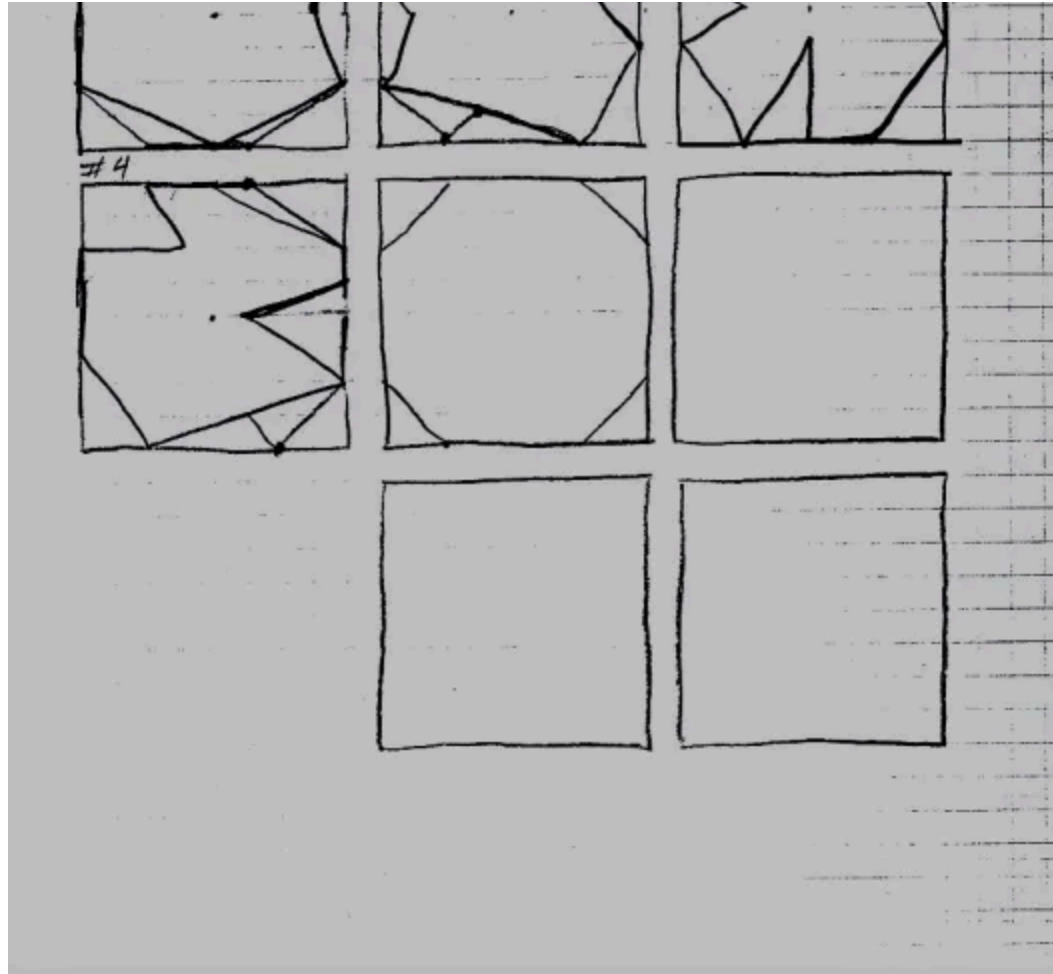
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screen, but a wrap-around warp mechanic was employed that meant that anything moving off the edge of the screen, reappeared at the opposite side to continue its path. The only way to deal with the asteroids, was to shoot them. The danger they represented would not simply float away. And so Logg found that early players of his code (usually fellow developers and engineers at the Atari plant) developed different ways of attacking the world. Some would shoot everything they could, and others were more circumspect about handling the floating rocks – either avoiding them altogether, or picking them off and breaking them down one at a time.

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Ed Logg's original hand drawn sketches of the rocks in Asteroids. (Courtesy of Ed Logg)

Observing the way his colleagues were approaching his game, Logg decided he had to push the player further to move

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bigger rocks out there, because more stuff on-screen means more chance of an unfortunate collision.

Enter the saucers:

Two saucers were developed and dropped into the game – one dumb, one smart. A large lumbering one that fired shots off randomly – handy for dashing player plans for neat Asteroids fields, as shots could easily hit passing rocks – and a smaller saucer – harder to hit, with built-in deadly accuracy with shots aimed at the player's ship. As described here:

I always wanted two saucers. A big one that fired randomly like cannon fodder to get you used to the concept that when you got down to fewer rocks, a ship was going to come out. The little saucer was about making you move. Run away, you're going to die if you stick around!

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212-688-6445

January 19, 1981

Mr. Edward Logg
Atari, Inc.
200 Fifth Avenue
New York, N.Y. 10010

Dear Mr. Logg:

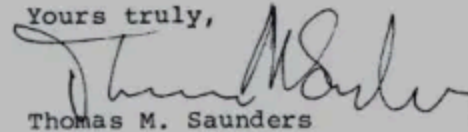
I was quite interested in reading the article entitled
"How to Win at Asteroids" in the February 1981 Esquire.

As the attorney for Walter Williams and Real Good Productions,
Inc., proprietors in all rights in the "Mr. Bill Show" and
related characters, I am always interested in public mention
of our names.

However, please be aware that the names "Mr. Bill" and
"Sluggo" are protected by United States trademark and copy-
right laws and various state and local laws.

I am sure the personalities associated with these names are
helpful in conceptualizing the behavior of elements of the
Asteroids game. Kindly refrain, however, from utilizing
these names in any phase of the manufacture, use or sale
of Asteroids.

Yours truly,



Thomas M. Saunders

TMS:jv

*copy was given to Skip Paul
a return statement.*

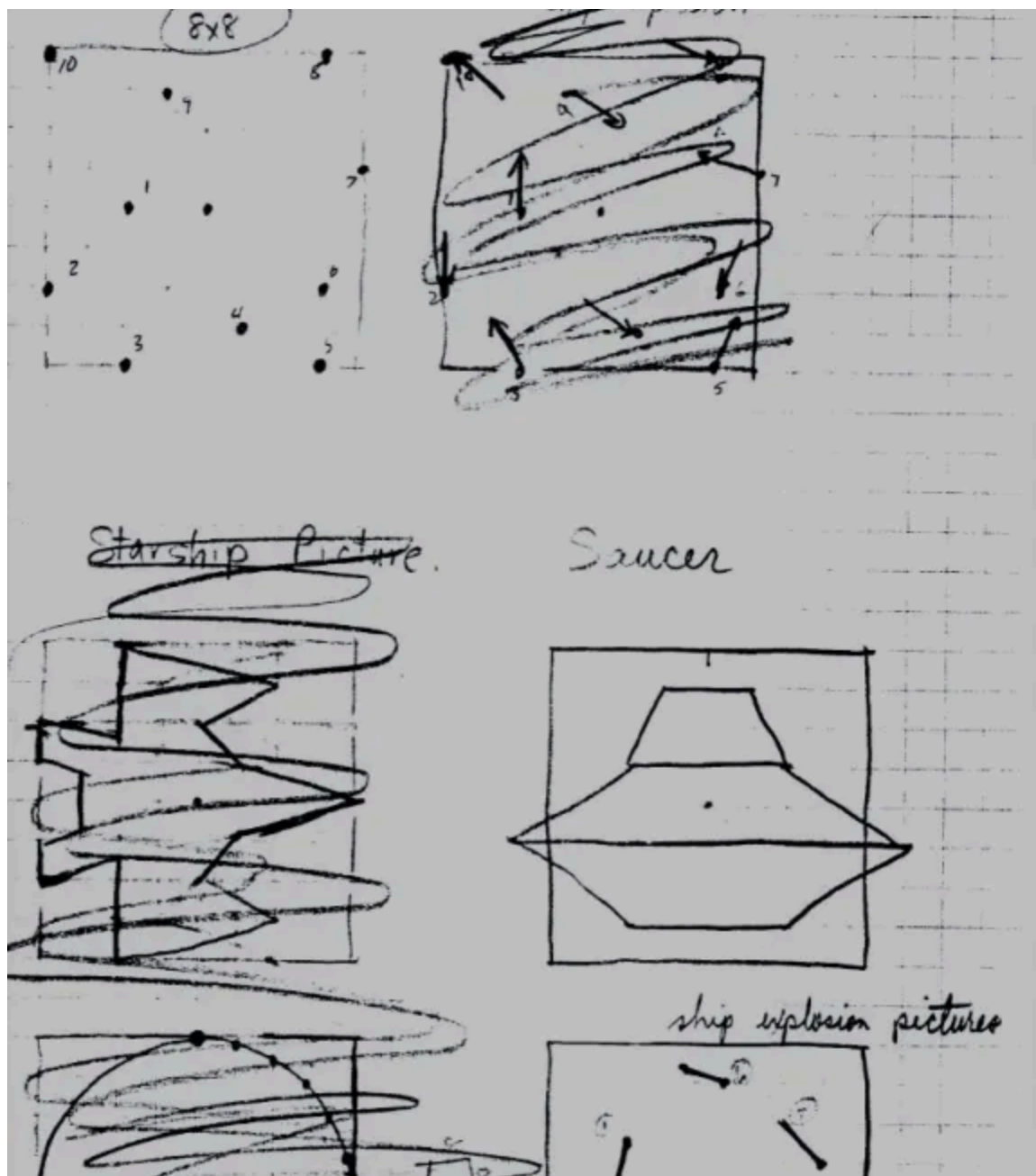
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Over the years, Logg has expressed two things that with hindsight, he would change to the final game. The Hyperspace button, and the behaviour of the small saucer:

The Hyperspace button is located some distance away from the other four control buttons. Hitting it will make your ship disappear and reappear a few moments later somewhere else on the screen, with a one in six chance of your ship exploding on re-entry. A neat feature intended to use as a last resort when death is inevitable, but one that many players never use, mainly because of how difficult it is to reach for the button!

The small saucer originally was programmed to shoot directly at the player as soon as it appeared on-screen. Field testing of the game gave feedback that this was unfair, so Logg changed the saucer's algorithm to miss the player with its first shot, to give a chance to respond. This led to the (in)famous "lurking" tactic used by players to this day, where a single rock can be left on-screen, and players can pick off 1,000 points a time by shooting down saucers as they appear on screen.

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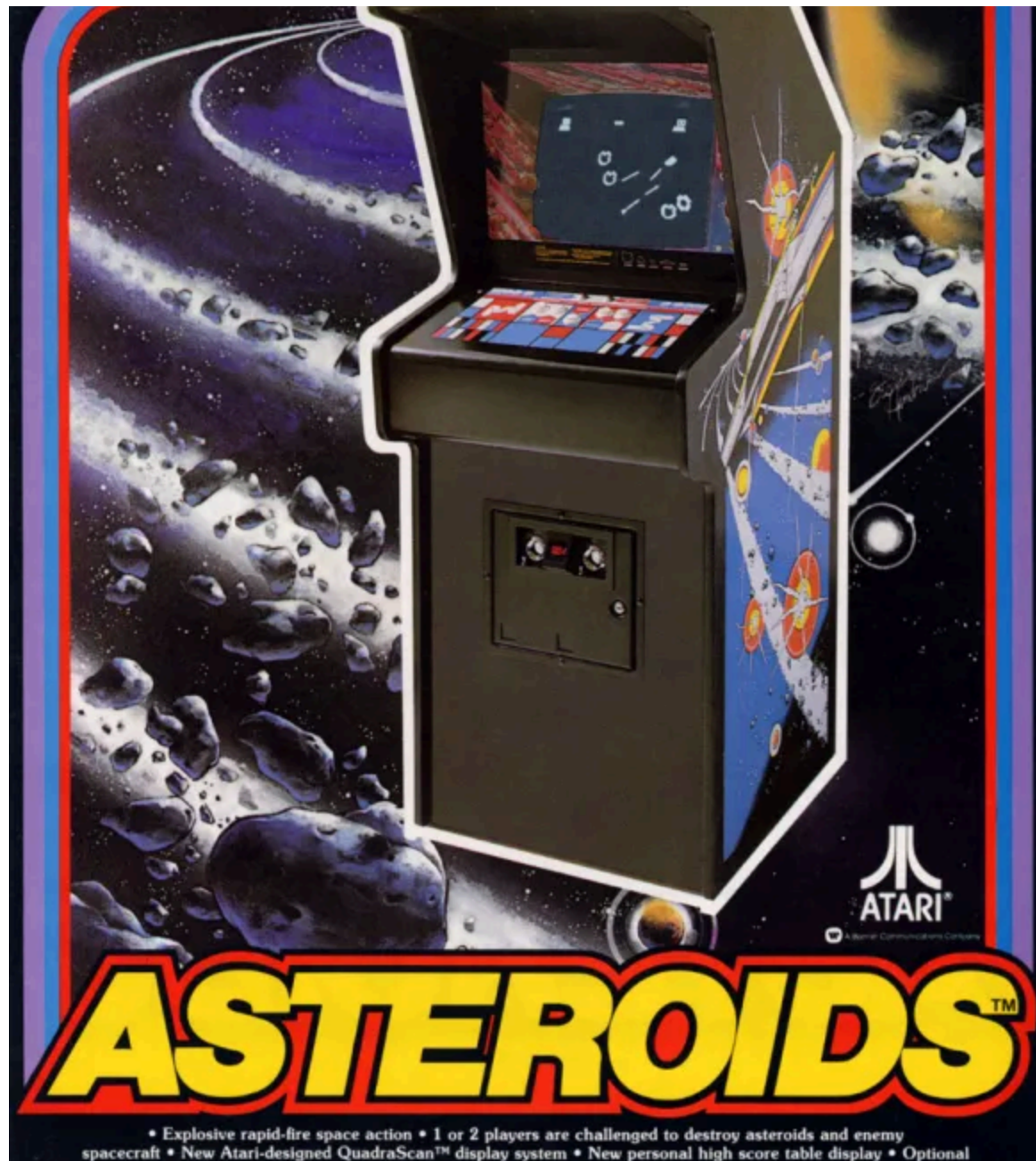
Ed Logg's design work on the ship explosion animation, a discarded idea for the player's ship, and saucer design. (Courtesy of Ed Logg)

Field Testing was done with two groups of players, both young and old. The feedback was very positive. [Take a look here at the original field test report.](#) (Courtesy of Ed Logg). Logg was inspired:

When we field-tested Asteroids for the very first time, I saw a player start a game and die three times; each within 20 seconds. He proceeded to put another quarter in. This tells me the player felt it was his fault he died and he was convinced he could do better. This is one of the primary goals a game designer tries to achieve and it was clear to me Asteroids had it.

More confirmation that the game was going to be a success.

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interesting insight to the reaction to the game within the company.

Asteroids is the only game that ever stopped production lines in our plant. At break time, the entire assembly line would run over to play the machines that were ready to be shipped out. With other games, the guys would just assemble them and box them up, and that was that. But with Asteroids, nobody wanted to work.

[This interesting document](#) details the raw cost to Atari of building an upright cabinet.

Logg knew he had created a hit. The game had a massive impact on Atari's fortunes. Outselling everything released by Atari to date, *Asteroids* quickly became part of the furniture not only in arcades, but in other locations too. Bars, airports, shopping malls and waiting rooms. Operators of buildings where large amounts of footfall were present, clamoured for this new game. Reports came in of some machines taking over \$1,000 a week in earnings, allowing the capital investment in the game to be repaid in a matter of days.

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Kids crowded around an Asteroids upright cabinet, circa 1980

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lured by the vector glow of the game.

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COIN CONNECTION

ATARI INC., 1265 BORREGAS AVENUE, SUNNYVALE, CALIFORNIA 94086

Operators Tell Us

Route operators report stronger video performance compared to pinball.

As we observe the skyrocketing sales and earning power of Atari's ASTEROIDS™, we are seeing a phenomenon that is becoming increasingly common to the coin-operated games market—the continuously high earning video game.

More and more route operators are carefully evaluating their mix of games in each location, and realizing that the video game no longer is second to the pinball game. Concerned about maximizing the profit in each location, operators are reassessing the video game, as the price difference between the two types of games becomes non-existent and the difference in the average earnings life disappears.

At the AMOA trade show in November 1979, over 450 operators participated in the first TELLUS SURVEY. Over 125 of these operators were interviewed in depth by teams of professional interviewers. Operators responded to questions about their business and buying behavior.

There was high representation of route and arcade operators who operate over 500 units. While video traditionally performs strongly in game centers and arcades, route operators revealed that video actually performs similarly or outperforms (according to some variables) when compared to pinball.

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New Asteroids™ Tops Production Records



Sunnyvale, CA. "To date we've produced more 'Asteroids' than any other game in our history, and orders are still piling up," reported Don Osborne, Atari National Sales Manager.

"It's our biggest hit yet, surpassing even some of our classic hits like 'Tank', 'Breakout', and 'Atari Football'."

"The unusually high response to this game is a testimonial to the fact that it is returning extremely high profits to operators all over the world. Asteroids has the kind of play appeal and reliability that gives it good location versatility. It continues to be highly successful in game rooms as well as street locations of every type," he said.

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VOLUME 4 NUMBER 2

FEBRUARY 1980

February 1980's edition of Atari's Coin Connection: a monthly newsletter for arcade operators and distributors reporting the success of Asteroids

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The late Don Osbourne, Atari's Sales Director, stated in 1980, that *Asteroids* at the height of its popularity "easily took \$10 million a week in quarters", an astonishing statistic if you think about it.

Ultimately, three cabinet types were released, in upright, cabaret and cocktail formats.

The game's success spawned an inevitable sequel, *Asteroids Deluxe*. Keeping the original mechanics, *Deluxe* saw the action sped up, and new features added – it was another huge hit for Atari.

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The follow-up to Asteroids, Asteroids Deluxe seen here in production at Atari's Tipperary plant in Ireland (Photo: Mike Jang)

Logg's legacy at Atari continued with the creation of *Centipede* and *Gauntlet* – both huge games in their own rights – but *Asteroids* was where things really took off. Logg's uncanny ability to hone gameplay with just the right balance of risk and reward, played into Atari's mantra of "Easy to learn, difficult to master", and earned him the nickname "Golden Boy" within the corridors of Atari.

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'Golden Boy' Ed Logg, with the Gold Asteroids, produced to celebrate 50,000 units built. It is believed this cabinet now sits at the Midway/Warner Bros offices in Chicago.

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What I learned from Ed was that creating the fun of a game did not require complex algorithms as much as it required the right approach. Which is to say that it wasn't virtuoso coding that made Ed's games a success, as much as it was putting all the proper features in the game in the correct order.

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Ed photographed in 2012 with an Asteroids upright cabinet.
Photo by [Gregg Segal](#)

Today, *Asteroids* remains at the heart of any serious arcade enthusiast's collection. I own a cabaret version of the game (with a multi-game kit, allowing play of *Asteroids*, *Asteroids Deluxe* and *Lunar Lander*). If you want to read the restoration

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social media platform.

See you next time.

Tony

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NEIL1637 says:

October 26, 2018 at 22:05

What a truly great write up. Fantastic information here Tony with some fabulous original documents and insights.

I should get my hands on my recently acquired cocktail cabinet next month, which won't work, but will be fun to restore.

Loading...



PHILIP CLARK says:

October 26, 2018 at 22:32

Do Lunar Lander!!!

Loading...



TONY says:

October 26, 2018 at 22:34

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JOHN says:

October 26, 2018 at 23:50

So great to read a fine treatment of these vector games I loved to play as a kid. Another one which I've hardly seen mentioned (after battle zone? Green tanks) is a ww1 aircraft vector game – I can't remember if it was tail gunner or red baron....

Loading...



CHUCK says:

October 27, 2018 at 04:15

What a great write-up Tony! I played Asteroids at Fun Spot (thanks to your suggestion!) a few weeks ago and can attest to just how amazing that phosphor glow is on the game. It is amazing how I played that decades ago and only now realize just how enchanting the vector monitor makes these games. The shots nearly jump off the screen, they are so bright.

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MIKE E says:

October 27, 2018 at 16:12

@John – Both Red Baron and Tail Gunner were vector games, but Red Baron would be the WW 1 game you would be thinking of.

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Pingback: Behind The Creation Of Atari's ,Asteroids,' An Arcade Classic | Tropische Reisen – News



MIKE JANG says:

October 28, 2018 at 19:25

Tony,
Great write up!
Mike

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JON RANDY says:

October 29, 2018 at 06:05

I have to say, the 'cabaret' version of the game that you have sounds intriguing!

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Pingback: [=== popurls.com === popular today](#)

Pingback: [Así fue la historia de cómo se creó "Asteroids" | TeraGames](#)

Pingback: [Atari Asteroids: Creating a Vector Arcade Classic – The Arcade Blogger – Datum](#)



GRIZZLYBEARFINANCE says:

October 30, 2018 at 13:40

I love that they had a form you had to fill out for a game proposal that was only big

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BILLY says:

October 30, 2018 at 21:08

The popping in another quarter after losing 3 lives in the first minute is so true. When I played it for the very first time, I had no idea how the game worked, but eventually got the hang of it. Thanks for the trip down memory lane.

Loading...



CAPTAIN OF HMS CQ says:

October 30, 2018 at 22:30

This might be the most nostalgic blog I've ever read. It takes me back to my first kiss (I meant Quarter, same dif...). Thank you!!!

Loading...



FLINNSTER says:

October 30, 2018 at 22:36

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Loading...



TONY says:

October 30, 2018 at 22:37

Flinnster, I have it. Will add it to the article when I get a minute.

Loading...



LESLIEJOCHASE says:

October 31, 2018 at 16:17

Your blog appeared on my WordPress Reader this morning so I started to read it and became fascinated. I had Seinfeld on in the background and, ironically, it was the episode "The Frogger!"

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STEVEBM1 says:

November 6, 2018 at 14:41

I always thought the artwork on the cab, looked a bit like cylon raiders from the original battlestar galactica, not sure if this was intentional or not

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Pingback: Gamasutra: Simon Carless's Blog – Video Game Deep Cuts: A Quiet Man, With Trivial Credits – Gamericans

Pingback: Gamasutra: Simon Carless's Blog – Video Game Deep Cuts: A Quiet Man, With Trivial Credits – Dinx.in

Pingback: Video Game Deep Cuts: A Quiet Man, With Trivial Credits – My Blog

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Pingback: Golden Age Arcade Games: Bootlegs, Lawsuits & Litigation – The Arcade Blogger



A31FORD says:

March 8, 2019 at 01:08

GREAT writeup, however, why is there no mention of the cabaret series 2 player machine (which I am a very proud owner of one.

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TONY says:

March 8, 2019 at 07:44

A two player cabaret? Tell us more!

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GREG MANNING says:

March 8, 2019 at 19:32

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for some 22 years now, and just recently when I went to power it up, did I find it now had small issues).

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TONY says:

March 8, 2019 at 19:47

Ah – a cocktail table. Yeah they are nice things to work on Greg. I like the wood grain on those.

Loading...



GREG MANNING says:

March 8, 2019 at 19:51

Just looked at what you call a cabaret version, assuming you are correct, that would make my version a cocktail version (table version) I had included a picture of mine in my last post.....

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Pingback: Script and Sources for Retro Release EP 24 – Asteroids – willnelson2508

Pingback: Arcade Heroes Newsbytes: More From EAG 2023; TRON 3; Space Invaders 45th; & More - Arcade Heroes

Pingback: Tech Time Warp: Take a nostalgia trip with the World Video Game Hall of Fame – Sabtrax



DAN S FORD says:

September 16, 2026 at 16:37

Excellent! The first game I purchased and still have.

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